

Convexity Adjustment Calculation

Interest Rate Futures

Convexity is an attribute of an interest rate instrument which measures how the instruments' duration or sensitivity to rate movements (i.e. DV01) changes as rate levels change. Convexity is often referred to as the second derivative of price with respect to rate level. A Convexity adjustment is a general concept important in many Fixed-Income derivative models whereby the rate of the instrument is adjusted to reflect the applicable convexity for that instrument. The definition of this adjustment varies depending on the context of the derivative instrument and there is no one standard model.

Most interest rate futures, such as Eurodollars, trade at a fixed value per basis point and thus have zero convexity. In contrast, a forward rate agreement or a swap containing a fixed-rate leg exhibits positive convexity similar to that of a bullet bond. Therefore, the rates implied by futures contracts are considered "expected rates" whereas FRA's and swaps are consistent with executable forwards, which encompass convexity.

The purpose of this document is to describe convexity adjustment whereby the interest rate future is converted into an equivalent forward rate agreement (FRA). This adjustment generates a lower rate for the futures contract, and becomes more pronounced the longer the expiry date of the futures contract.

Convexity Correction Formulas

From a quantitative point of view the origin of the convexity correction can be expressed as the fact that futures and forwards are expected values of the forward rate but under different measures. The futures rate, due to the daily margining feature is the expected value of the forward rate under the risk neutral measure. This differs from the forward rate and is a model dependent quantity, that is, it depends on the dynamics of the interest rate. A common approach to compute convexity corrections is to use so called short rate models for the interest rate evolution. A popular choice corresponds to the use of the Hull-White model which postulates that the short rate under the risk neutral measure follows a normal diffusion process with volatility σ and speed of mean reversion a .

The relation between (simple compounded) forward rate $fwd_0(T_1, T_2)$ from T_1 to T_2 and the observable (simple compounded) future rate $fut_0(T_1, T_2)$ over the same period is:

$$fwd_0(T_1, T_2) \approx fut_0(T_1, T_2) - [1 - \exp(-X(T_1, T_2))] \left[fut_0(T_1, T_2) + \frac{1}{\delta} \right]$$

where

$$\begin{aligned} \delta &= T_2 - T_1 \\ X(T_1, T_2) &= \frac{\sigma^2}{2a} B(T_1, T_2) (B(T_1, T_2)(1 - e^{-2aT_1}) + aB^2(0, T_1)) \\ B(t, T) &= \frac{1 - e^{-a(T-t)}}{a} \end{aligned}$$

Derivation of this formula is provided in the appendix.

Convexity Adjustment Settings

Mean Reversion Speed

The mean reversion speed a represents the speed with which the short term rate reverts to its long term mean. The typical range of values for mean reversion is 0.001 for negligible effects to 0.1, which is a relatively high mean reversion. Its inverse $1/a$ is the time scale on which mean reversion takes place.

Rate Volatility

The value used as default for the short rate volatility σ is chosen as the approximate normal (Bachelier) 10 year cap volatility obtained by multiplying the 10 year Black volatility by the corresponding swap rate.

$$\text{Rate volatility} = (\text{10-yr swap rate } \{\text{USSWAP10}\}) * (\text{10-yr cap volatility } \{\text{USCV10}\}) / 100$$

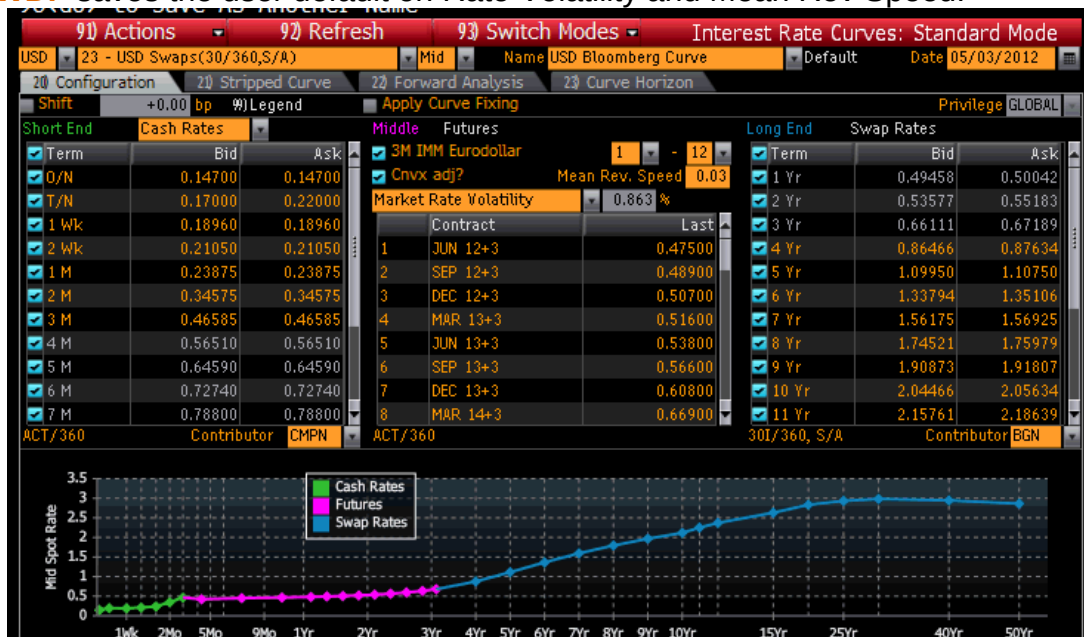
For each currency, different tenors may be used for rate volatility. The choice of which tenors is based on the unique set of market conditions (i.e. liquidity) for each swap curve and individual tenors. One recent market condition impacting determination of what market data to use is the continued existence of negative rates. As a result, IR VOL market players have been migrating to normal volatilities for trading; therefore, EUR and CHF was recently enhanced to use normal volatility. For other currencies (besides EUR and CHF) if the swap rate becomes negative then the forward swap rate and Swaption volatility is used for rate volatility.

Below we list several currencies and which tenors are used:

USD: USCV10 * USSWAP10
EUR: EUNS06 BVOL
GBP: BPNS06 BVOL
JPY: JYCV5 * JYSWAP5
CHF: SFNS03 BBIR
CAD: CDCV03* CDSW3
AUD: ADCV5* ADSWAP5
SEK: SKSN055 BBIR
SGD: SDCV2* SDSW2
HKD: HDCV2* HDSW2
NZD: NDSN011 BBIR

Bloomberg Screens

SWDF saves the user default on Rate Volatility and Mean Rev Speed.



EDSF and **EDS** always assumes 0.03 as default for Mean Rev Speed, and use cap volatility for the currency.

United States		97 Settings	98 Feedback	Synthetic Forward Rates							
Delayed	05/03/2012	Adjust	Mean Rev.	0.030	Rate Vol.	0.863%	9) Legend				
Start Date	Days	Ticker	Last	Rate	6 Mo	1 Yr	2 Yr	3 Yr	5 Yr	7 Yr	10 Yr
5/7/2012	44	Front	99.7149	0.2851	0.434	0.471	0.537	0.672	1.161	1.682	2.244
10) 6/20/2012	91	EDM2	99.5250	0.4750	0.483	0.500	0.561	0.708	1.215	1.733	2.283
11) 9/19/2012	91	EDU2	99.5100	0.4900	0.500	0.517	0.600	0.784	1.323	1.837	
12) 12/19/2012	91	EDZ2	99.4900	0.5100	0.515	0.539	0.653	0.873	1.438	1.941	
13) 3/20/2013	91	EDH3	99.4800	0.5200	0.533	0.566	0.722	0.975	1.557	2.046	
14) 6/19/2013	91	EDM3	99.4550	0.5450	0.560	0.608	0.809	1.088	1.681	2.152	
15) 9/18/2013	91	EDU3	99.4250	0.5750	0.598	0.668	0.915	1.218	1.808	2.258	
16) 12/18/2013	91	EDZ3	99.3800	0.6200	0.653	0.751	1.037	1.353	1.939	2.365	
17) 3/19/2014	91	EDH4	99.3150	0.6850	0.736	0.860	1.176	1.493	2.071	2.471	
18) 6/18/2014	91	EDM4	99.2150	0.7850	0.846	0.991	1.325	1.650	2.202	2.575	
19) 9/17/2014	91	EDU4	99.0950	0.9050	0.981	1.140	1.487	1.804	2.331	2.677	
20) 12/17/2014	91	EDZ4	98.9450	1.0550	1.132	1.300	1.647	1.959	2.456	2.776	
21) 3/18/2015	91	EDH5	98.7950	1.2050	1.292	1.466	1.804	2.112	2.576	2.870	
22) 6/17/2015	91	EDM5	98.6250	1.3750	1.460	1.630	1.971	2.260	2.690	2.960	
Start Date	End Date	Days	Years	Front Stub	Days	Back Stub	Days	Bond Yield	M-Mkt Yield		
05/07/2012	05/07/2013	365	1.00	0.2851	44	0.5198	48	0.478	0.471		
05/07/2012	05/07/2014	730	2.00	0.2851	44	0.6847	49	0.537	0.531		
05/07/2012	05/07/2015	1095	3.00	0.2851	44	1.2042	50	0.672	0.668		
05/07/2012	05/07/2016	1461	4.00	0.2851	44	1.8531	52	0.898	0.899		
05/07/2012	05/07/2017	1826	5.00	0.2851	44	2.4712	53	1.161	1.175		

IMM EURODOLLAR FUTURES ANALYSIS										
5/ 3/12	Valuation	Mean Rev Speed 0.030			Rate Volatility 0.863%					
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Valuation	#of	Futures	Future	Adjust	Cash	Convexi	Future	Adj Fu	Cash	=
Date	Days	Price	Rate	Rate	Forwrd	Bias ^{bp}	Strip	Strip	Strip	8-9
6/20/12	48	99.5250	0.4750	0.4747	0.7771	0.03	0.3029	0.3029	0.3029	0.00 ^{bp}
9/19/12	139	99.5100	0.4900	0.4887	1.1125	0.13	0.4157	0.4155	0.6136	0.02
12/19/12	230	99.4900	0.5100	0.5073	1.3726	0.27	0.4454	0.4448	0.8120	0.06
3/20/13	321	99.4800	0.5200	0.5155	1.0532	0.45	0.4641	0.4629	0.9729	0.12
6/19/13	412	99.4600	0.5400	0.5332	0.2761	0.68	0.4832	0.4812	0.9927	0.20
9/18/13	503	99.4250	0.5750	0.5655	0.0235	0.95	0.4949	0.4921	0.8636	0.29
12/18/13	594	99.3800	0.6200	0.6074		1.26	0.5086	0.5047	0.7349	0.39
3/19/14	685	99.3150	0.6850	0.6689	0.1967	1.61	0.5246	0.5196	0.6066	0.51
6/18/14	776	99.2150	0.7850	0.7650	0.8423	2.00	0.5447	0.5383	0.5588	0.64
9/17/14						2.43	0.5712	0.5633	0.5896	0.79
EDS<go> select 1										
(4)-(5)=(7)										
Futures daytype: ACT/360										
Strip yield (<1 yr): ACT/360										
Strip/Coupn (>1 yr): bond equiv										
Hit <Menu> to return main page										
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For more information, please press the **<HELP>** key twice on the **BLOOMBERG PROFESSIONAL®** service.

Appendix: Derivation of Convexity Correction Formulas

In the Hull-White model with mean reversion a and constant volatility σ ,

$$dr(t) = (\theta(t) - a r(t)) dt + \sigma dW(t)$$

where $\theta(t)$ is chosen to fit the interest rate term structure.

The solution to this SDE is:

$$r(t) = \alpha(t) + \sigma \int_0^t e^{-a(t-u)} dW(u)$$

where

$$\alpha(t) = f^M(0, t) + \frac{\sigma^2}{2a^2} (1 - e^{-at})^2$$

and $f^M(0, t)$ is the market instantaneous forward rate at time 0 for the maturity T .

Some notation before proceeding to the derivation of convexity correction.

Denote the coverage period δ of the forward rate from T_1 to T_2 and the function $B(t, T)$ as:

$$\delta \stackrel{\text{def}}{=} T_2 - T_1$$

$$B(t, T) \stackrel{\text{def}}{=} \frac{1 - e^{-a(T-t)}}{a}$$

The time 0 value of a zero coupon bond maturity at T can be expressed as following under the risk neutral measure Q :

$$\begin{aligned} P(0, T) &= E^Q \left[\exp \left(- \int_0^T r(t) dt \right) \right] \\ &= E^Q \left[\exp \left(- \int_0^T \alpha(t) dt - \sigma \int_0^T \int_0^t e^{-a(t-u)} dW_u dt \right) \right] \\ &= \exp \left(- \int_0^T \alpha(t) dt \right) E^Q \left[\exp \left(- \sigma \int_0^T \int_0^t e^{-a(t-u)} dW_u dt \right) \right] \\ &= \exp \left(- \int_0^T \alpha(t) dt \right) E^Q \left[\exp \left(- \sigma \int_0^T \int_u^T e^{-a(t-u)} dt dW_u \right) \right] \\ &= \exp \left(- \int_0^T \alpha(t) dt \right) E^Q \left[\exp \left(- \sigma \int_0^T B(u, T) dW_u \right) \right] \\ &= \exp \left(- \int_0^T \alpha(t) dt + \frac{\sigma^2}{2} \int_0^T B^2(t, T) dt \right) \end{aligned}$$

The time zero (simple compounded) forward rate is related to the bond price as:

$$\begin{aligned} 1 + \delta * fwd_0(T_1, T_2) &= \frac{P(0, T_1)}{P(0, T_2)} \\ &= \exp\left(\int_{T_1}^{T_2} \alpha(t) dt + \frac{\sigma^2}{2} \int_0^{T_1} B^2(t, T_1) dt - \frac{\sigma^2}{2} \int_0^{T_2} B^2(t, T_2) dt\right) \end{aligned}$$

As described above, the time zero (simple compounded) future rate under the risk neutral measure is given by:

$$\begin{aligned} &1 + \delta * fut_0(T_1, T_2) \\ &= E^Q \left[\exp\left(\int_{T_1}^{T_2} r(t) dt\right) \right] \\ &= E^Q \left[\exp\left(\int_{T_1}^{T_2} \alpha(t) dt + \sigma \int_{T_1}^{T_2} \int_0^t e^{-a(t-u)} dW_u dt\right) \right] \\ &= \exp\left(\int_{T_1}^{T_2} \alpha(t) dt\right) E^Q \left[\exp\left(\sigma \int_{T_1}^{T_2} \int_0^t e^{-a(t-u)} dW_u dt\right) \right] \\ &= \exp\left(\int_{T_1}^{T_2} \alpha(t) dt\right) E^Q \left[\exp\left(\sigma \int_0^{T_1} \int_{T_1}^{T_2} e^{-a(t-u)} dt dW_u + \sigma \int_{T_1}^{T_2} \int_u^{T_2} e^{-a(t-u)} dt dW_u\right) \right] \\ &= \exp\left(\int_{T_1}^{T_2} \alpha(t) dt\right) E^Q \left[\exp\left(\sigma \int_0^{T_1} B(u, T_2) - B(u, T_1) dW_u + \sigma \int_{T_1}^{T_2} B(u, T_2) dW_u\right) \right] \\ &= \exp\left(\int_{T_1}^{T_2} \alpha(t) dt + \frac{\sigma^2}{2} \int_0^{T_1} (B(t, T_2) - B(t, T_1))^2 dt + \frac{\sigma^2}{2} \int_{T_1}^{T_2} B^2(t, T_2) dt\right) \end{aligned}$$

The ratio of the future rate and forward rate expressions is:

$$\begin{aligned} &\frac{1 + \delta * fut_0(T_1, T_2)}{1 + \delta * fwd_0(T_1, T_2)} \\ &= \exp\left(\frac{\sigma^2}{2} \left(\int_0^{T_1} (B(t, T_2) - B(t, T_1))^2 dt + \int_{T_1}^{T_2} B^2(t, T_2) dt - \int_0^{T_1} B^2(t, T_1) dt + \int_0^{T_2} B^2(t, T_2) dt \right)\right) \\ &= \exp\left(\sigma^2 \int_0^{T_1} B^2(t, T_2) - B(t, T_1)B(t, T_2) dt\right) \exp\left(\sigma^2 \int_{T_1}^{T_2} B^2(t, T_2) dt\right) \end{aligned}$$

We approximate this by dropping the second exponential $\exp\left(\sigma^2 \int_{T_1}^{T_2} B^2(t, T_2) dt\right)$. In the case of large T_1 , this term would be small compare to the term in the first exponential $\exp\left(\sigma^2 \int_0^{T_1} B^2(t, T_2) dt\right)$. In the case of small T_1 and T_2 , the overall convexity adjustment would be small anyway and the effect of omitting this term would be negligible. Thus, we have the following approximation:

$$\frac{1 + \delta * fut_0(T_1, T_2)}{1 + \delta * fwd_0(T_1, T_2)} \approx \exp\left(\sigma^2 \int_0^{T_1} B^2(t, T_2) - B(t, T_1)B(t, T_2) dt\right)$$

The expression inside this exponential function is equivalent to the following function:

$$\begin{aligned} X(T_1, T_2) &\stackrel{\text{def}}{=} \sigma^2 \int_0^{T_1} B^2(t, T_2) - B(t, T_1)B(t, T_2) dt \\ &= \frac{\sigma^2}{2a} B(T_1, T_2) (B(T_1, T_2)(1 - e^{-2aT_1}) + aB^2(0, T_1)) \end{aligned}$$

With further arrangement of above equation, we have the following relationship between the (simple compounded) forward rate and the observable (simple compounded) future rate:

$$\begin{aligned} \frac{1 + \delta * fut_0(T_1, T_2)}{1 + \delta * fwd_0(T_1, T_2)} &\approx \exp(X(T_1, T_2)) \\ fwd_0(T_1, T_2) &\approx fut_0(T_1, T_2) - [1 - \exp(-X(T_1, T_2))] \left[fut_0(T_1, T_2) + \frac{1}{\delta} \right] \end{aligned}$$